Chronic Total Occlusion PCI Techniques and Outcomes

Clinical Consensus Document · Part 32

Background and Rationale:

Retrograde and antegrade approaches for Chronic Total Occlusion (CTO) crossing. Wire escalation algorithms, dissection and re-entry techniques (CrossBoss and Stingray), and microcatheter selection.

Coronary artery disease involving complex anatomical subsets presents significant challenges. Technological advances including Intravascular Ultrasound (IVUS), Optical Coherence Tomography (OCT), Excimer Laser Coronary Atherectomy (ELCA), and Intravascular Lithotripsy (IVL) have revolutionized lesion preparation and stent implantation.

Indication Criteria for CTO:

1. Hemodynamic instability or ongoing ischemia with FFR < 0.80 or resting iFR < 0.89.
2. Severe calcification precluding stent delivery: recommended upfront treatment with IVL or ELCA.
3. In-Stent Restenosis (ISR): comprehensive assessment for under-expansion or neoatherosclerosis via OCT.
4. Left Main (LM) bifurcation disease requiring dedicated provisional or dual-stent strategy.

Section 2: Technical Specifications and Evidence Review

Detailed Procedural Steps (Protocol Step 1):

Radial access with 6F or 7F guide catheter is standard. Anticoagulation achieved with unfractionated heparin targeting an activated clotting time (ACT) of 250-300 seconds. Guide catheter extension (GuideLiner) is recommended for deep vessel support in tortuous calcified anatomy.

Stent sizing must adhere to distal reference vessel external elastic lamina (EEL) measured by IVUS. Post-dilation with high pressure non-compliant balloon (18-24 atm) is mandatory to prevent acute stent thrombosis.

Table 1.1 ? Nominal Pressures and Burst Pressures

Stent Diameter (mm)	Nominal (atm)	RBP (atm)	Target Minimum Area (mm²)
2.50 mm	9.0 atm	16.0 atm	4.5 mm²
3.00 mm	9.0 atm	18.0 atm	5.5 mm²
3.50 mm	10.0 atm	18.0 atm	7.0 mm²
4.00 mm (LM)	10.0 atm	20.0 atm	8.5 mm²

RBP: Rated Burst Pressure; LM: Left Main.

Section 3: Technical Specifications and Evidence Review

Detailed Procedural Steps (Protocol Step 2):

Radial access with 6F or 7F guide catheter is standard. Anticoagulation achieved with unfractionated heparin targeting an activated clotting time (ACT) of 250-300 seconds. Guide catheter extension (GuideLiner) is recommended for deep vessel support in tortuous calcified anatomy.

Stent sizing must adhere to distal reference vessel external elastic lamina (EEL) measured by IVUS. Post-dilation with high pressure non-compliant balloon (18-24 atm) is mandatory to prevent acute stent thrombosis.

Table 2.1 ? Nominal Pressures and Burst Pressures

Stent Diameter (mm)	Nominal (atm)	RBP (atm)	Target Minimum Area (mm²)
2.50 mm	9.0 atm	16.0 atm	4.5 mm²
3.00 mm	9.0 atm	18.0 atm	5.5 mm²
3.50 mm	10.0 atm	18.0 atm	7.0 mm²
4.00 mm (LM)	10.0 atm	20.0 atm	8.5 mm²

RBP: Rated Burst Pressure; LM: Left Main.